# When Innovation Is Too Fast: Reading, Writing, and the Crowding of Science

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**Abstract**

Science runs on writing new results and reading other people's. Reading *teaches*: it raises the quality of new work. It also *protects*: it reveals projects that others have already done. We model researchers who choose a topic and split their time between reading and writing. The payoff of working in a topic is single peaked in the flow of its literature: thin topics offer little to learn from, dense ones hide more unread overlap. Free choice of topics then has three consequences. The gains from cumulative knowledge are competed away. The overlap rate is pinned by entry, not by volume. And at any stable equilibrium the marginal paper hurts its own topic. A catch-up cost, the need to read a topic's stock before contributing, protects incumbents' rents. As a result, measured returns rise with density even though topics are too crowded. Topics whose literature grows faster than a threshold rate overshoot their best density: innovation is "too fast". Faster writing scales science; better reading crowds it. Pre-registered tests on 1.7 million arXiv papers support the core mechanism and the catch-up friction. We calibrate the model and derive a sufficient-statistic formula for the stepping-on-toes parameter of growth theory.

**JEL:** O31, O33, O40, D62, I23. **Keywords:** economics of science, rent dissipation, topic choice, duplication, burden of knowledge, artificial intelligence.

*Draft of October 2026 for the project *When Innovations Are Too Fast*. The arXiv evidence summarised in Section 2 comes from the project's pre-registered tests. Code that reproduces every number, figure and numerical check in this paper is in the accompanying repository (`code/`).

# 1 Introduction

Submissions to arXiv more than doubled between 2018 and 2025, and large language models now draft text faster than any researcher can. A natural worry follows: innovation may be produced faster than it can be absorbed. When the literature grows faster than people can read it, researchers miss work that covers their own, duplicate it, and build less on what came before. On this view, cheaper writing is a mixed blessing and better reading tools are the remedy.

This paper argues that the worry and the remedy both misread how science adjusts. Researchers choose not only how much to write but *where*. When the literature grows, they spread out. The rate at which they duplicate each other is set by that choice of where to work, not by the volume of papers. The cost of fast innovation therefore does not show up as more duplication. It shows up in three less visible places. The first is the gains from building on others' work, which crowding competes away. The second is a catch-up burden that lets incumbents keep rents and hides overcrowding behind rising returns. The third is a class of hot topics whose literature grows faster than newcomers can catch up, and which overshoot their best density.

**The model.** Researchers choose a topic and split their time between writing and reading. Reading has two jobs. It *teaches*: what one reads raises the quality of one's own work, more so in a topic with more to learn from. It *protects*: each earlier paper may cover one's project, and reading is how one finds out. Unaware duplicates lose part of their credit. A detected overlap can be turned into a follow-on that cites the earlier paper, or abandoned at a cost. Topics can be joined or opened, and opening one is a lottery whose odds worsen as fruitful directions are taken. Newcomers to a topic must first read its stock of literature, the topic-level analogue of the burden of knowledge of Jones (2009).

**Results.** We prove seven sets of results.

(1) *The payoff curve is single peaked* (Theorem 1). The payoff of working in a topic first rises with the topic's flow of papers, because there is more to learn from, and then falls, because more covering papers go unread. This holds for any learning and detection technology without increasing returns to reading and with diminishing returns to search. The proof rests on joint log-concavity of the researcher's problem in time and density.

(2) *The sign of the spillover is derived* (Proposition 1). By the envelope theorem, an extra paper changes each incumbent's payoff by the slope of the payoff curve. It helps thin topics and hurts dense ones.

(3) *Dissipation and pinned overlap* (Propositions 2 and 3). With free choice, topics fill past the peak until they are no better than the frontier, so the gains from cumulative knowledge are competed away. Equilibrium density responds to the size of the community with an elasticity $\mathcal{E} = \eta/(\eta D + |\varepsilon_w|)$ that vanishes as the frontier becomes elastic. Volume opens new topics instead of thickening old ones, and the overlap rate is pinned by entry, not by volume.

(4) *Rents and overcrowding* (Propositions 5 and 6). Catch-up costs protect incumbents' rents, so returns rise with density across topics. The returns gradient equals the catch-up premium. Yet at any stable equilibrium the marginal newcomer *lowers* total output in her topic once the reading she imposes on future newcomers is counted. Rising returns coexist with overcrowding.

(5) *Too fast* (Proposition 7). Because the stock lags the flow, a topic whose literature grows faster than the threshold $g^* = \rho\hat{P}/(b_c \ln(\hat{v}/F)) - \delta$ keeps attracting newcomers past its peak, and stays overcrowded long after the rush. Faster catch-up tools *lower* the threshold.

(6) *Writing AI scales science; reading AI crowds it* (Propositions 8 and 10). Faster drafting multiplies output almost one for one and leaves duplication unchanged. Better reading, search or catch-up tools raise density and shrink the number of topics, and with an elastic frontier they leave output per researcher unchanged: a Solow paradox for science.

(7) *Stepping on toes, measured* (Proposition 9). The parameter $\lambda$ of Jones (1995) equals $1 - \mathcal{E}|\varepsilon_w|$, the product of the volume elasticity of density and the slope of the entrant's payoff curve. Crowding buffers frontier depletion rather than amplifying it.

The planner opens more topics and puts each at the peak of the entrant's payoff curve, where the writing wedge vanishes. A frontier subsidy decentralises the optimum. Policies that act only inside topics are dissipated.

**Evidence.** We confront the model with seven facts from pre-registered tests on 1.71 million arXiv papers (Section 2). The topic-choice core survives its sharpest test. When an area's literature grows, the space around a typical paper fills up with an elasticity of 0.04–0.27, against 0.53–0.89 if papers kept landing in the same idea space. Full dissipation fails. The same researcher earns more in denser topics up to about one standard deviation above the mean, and less beyond it. The model's catch-up cost reproduces this hump: mature topics sit just below the peak, protected by the catch-up premium, and fast-growing topics overshoot it. Unaware duplicates lose about a fifth of their citations and aware follow-ons gain about 30%, which supports separating the two kinds of overlap with partial credit. The AI channels can be told apart. Flat papers per author, more pioneer papers where LLMs spread, a flat aware share and flat overlap measured by meaning all say that AI has so far acted on science through entry, not through reading.

**Quantification.** The calibrated model (Section 9) puts the gain from cumulative knowledge that crowding dissipates at 18% of output per researcher, and catch-up at 1.9 years of a newcomer's career. Topics growing faster than 22% a year overshoot. The volume elasticity of 0.2 implies a frontier elasticity of 0.042 and a stepping-on-toes parameter of 0.968. Topic crowding therefore accounts for little of the measured decline in research productivity (Bloom et al., 2020), and the rest must come from falling fertility of new ideas or from selection.

**Related literature.** The paper joins four literatures. The first is the economics of science and priority (Merton, 1957, 1961; Dasgupta and David, 1994; Stephan, 1996; Hill and Stein, 2025). We add reading as a choice, with two distinct roles, and derive rather than assume the costs of duplication. The second is the theory of common-pool resources and excess entry (Gordon, 1954; Weitzman, 1974; Mankiw and Whinston, 1986) and of patent races (Loury, 1979; Dasgupta and Stiglitz, 1980). Our resource is attention to a literature, and the crowders are the readers. This makes the sign of the externality endogenous and lets catch-up costs turn dissipated rents into protected ones. The third is the direction of research (Bryan and Lemus, 2017; Hopenhayn and Squintani, 2021; Bramoullé and Saint-Paul, 2010; Rzhetsky et al., 2015; Foster et al., 2015; Myers, 2020; Borjas and Doran, 2012; Azoulay et al., 2019). We locate the inefficiency in the density of topics rather than in their riskiness, and derive testable signatures from the arXiv record. The fourth is the macroeconomics of ideas (Romer, 1990; Aghion and Howitt, 1992; Jones, 1995; Kortum, 1997; Weitzman, 1998; Bloom et al., 2020; Jones, 2009) and of access to knowledge (Furman and Stern, 2011; Murray et al., 2016; Iaria et al., 2018; Biasi and Moser, 2021; Evans, 2008; McCabe and Snyder, 2015; Chu and Evans, 2021; Park et al., 2023). We give stepping on toes an equilibrium foundation with observable inputs and explain why improvements in access to the literature need not raise research productivity. On artificial intelligence and science (Agrawal et al., 2019; Brynjolfsson et al., 2019; Hao et al., 2026; Liang et al., 2024), we separate writing, entry, reading, search and catch-up channels and show which ones the data reveal.

Section 2 presents the facts. Section 3 sets up the model. Section 4 derives the payoff curve. Sections 5 and 6 characterise equilibrium without and with catch-up costs. Sections 7 and 8 study AI and aggregate implications. Section 9 quantifies the model, Section 10 lists held-out predictions, and Section 11 concludes.

## 2 Seven facts from arXiv

The evidence comes from the project's pre-registered tests on 1.71 million arXiv papers posted 2018–2025, classified into 150 text-based fine areas. These are complemented by a PDF sample of 25,400 papers for citation behaviour and OpenAlex citation counts. A paper's *overlap flag* is set when an earlier paper by another group, posted 3 to 24 months before, lies above its area's 99th percentile of 2018–19 nearest-neighbour similarity. The *level* of the flagged rate is therefore a normalisation, about 1% in the base period. Its *changes* carry the information. Every prediction tested below was written down before the test was run. Table 1 collects the numbers.

**Fact 1.** *Volume opens new ground.* The elasticity of the local-neighbour hazard with respect to an area's prior two-year volume is 0.04–0.19 across similarity radii, with area and year effects. If an area's papers kept landing in the same idea space, randomly thinning each paper's observed neighbour list implies 0.53–0.89. In the all-arXiv time series, volume rose 113% from 2018 to 2025 while the overlap hazard rose 13% (elasticity 0.19–0.27). Close overlap responds least of all.

Table 1: Targets and tests from the arXiv evidence

| Fact | Statistic | Value |
|---|---|---|
| 1 | Elasticity of local density to area volume | 0.04–0.19 (time series 0.19–0.27) |
| | Fixed-field benchmark | 0.53–0.89 |
| 2 | Papers per author while authors grew | flat |
| 3 | Citations per author-year per SD of density (author FE) | +6.7% (SE 0.15) |
| | By fifth, relative to middle: sparsest / 4th / densest | −8% / +5% / +1% |
| | Peak of quadratic; share of researcher-years beyond it | ≈ +1 SD; 12% |
| 4 | Citations: unaware overlap / aware overlap | −19% (2) / +29% (5) |
| 5 | Aware share of close overlap, 2019–2026 | 35–42%, flat |
| 6 | Pioneer papers vs. trend, high-uptake areas, 2025H2+ | +1.3 pp (0.5) |
| 7 | Close overlap by meaning, high- vs. low-uptake, 2024–26H1 | −0.19 pp (0.24) |

Notes: arXiv 2018–2025, 150 areas. Standard errors in parentheses. Facts 1 and 3 discipline the calibration of Section 9. Facts 2 and 5–7 test the AI channels of Section 7. Fact 4 disciplines the credit block.

**Fact 2.** *Growth came from new authors.* Papers per author stayed flat while the number of authors grew.

**Fact 3.** *Returns are hump-shaped in topic density.* Comparing the same researcher across years spent in denser or sparser topics, with author and year effects (923,000 author-years), one standard deviation of density goes with 6.7% more fractional citations per author-year. By fifth of density, returns relative to the middle fifth are −8% in the sparsest fifth, rise to +5%, and fall back to +1% in the densest. A quadratic puts the peak about one standard deviation above mean density, with 12% of researcher-years beyond it. The shape survives using the third-closest paper and restricting to researchers active three or more years.

**Fact 4.** *Duplicates lose some credit, aware follow-ons gain.* Relative to papers in the same area-year with the same topic density, *unaware* overlaps (which do not cite their nearest earlier paper) receive 19% fewer citations. *Aware* overlaps (which cite it) receive 29% more.

**Fact 5.** *Awareness did not rise.* The aware share of close overlap stayed between 35% and 42% in every year from 2019 to 2026. It did not differ by exposure to large language models: −0.5 pp per standard deviation, SE 1.4.

**Fact 6.** *No concentration where LLMs spread.* High-uptake areas were already losing pioneer papers before ChatGPT. Against their own extended trend, they produced *more* pioneer papers afterwards: +0.4 pp (SE 0.3) in 2023–2025H1 and +1.3 pp (SE 0.5) from 2025H2. Local density fell by 1.1 and 2.7 pp.

**Fact 7.** *The 2026 drop in measured overlap is a change in wording.* Word-based (TF-IDF) overlap fell in 2026 across the whole similarity distribution, concentrated in high-uptake areas rather than dense ones. Measured by meaning (sentence embeddings), close overlap did not fall.

How do the facts map into the model? Fact 1 is the volume elasticity $\mathcal{E}$ of Proposition 3. Fact 3

rejects the equal returns of the frictionless model and motivates the catch-up cost of Section 6. Fact 4 disciplines the credit parameters $\lambda$ and $\Delta/\lambda$ (Lemma 1). Facts 2 and 5–7 distinguish the AI channels of Section 7.

# 3 Model

The economy is stationary. Time is continuous, but within a topic the relevant literature is a flow, so we describe a researcher's choices per unit of time.

## 3.1 Researchers, topics and time

There is a continuum of topics. Topics that are open are ex ante identical. A mass $N$ of researchers each has one unit of time per period, split into a share $x$ spent writing and a share $r = 1 - x$ spent reading. Writing full time yields $a$ papers (projects) per period. Topic $j$ hosts $n_j$ researchers who write, and receives a flow of

$$P_j = a\, x\, n_j \tag{1}$$

papers per period. $P_j$ is the topic's *density*. We call the ratio of writing speed to reading speed, $a/b$, the *pace* of science.

## 3.2 Reading teaches

A researcher who reads full time reads $b$ papers per period, so a reading share $r$ yields $R = b\, r$ papers drawn from her topic's flow. What she reads raises the quality of her own work by the factor $Q(R, P)$.

**Assumption 1** (Learning)**.** $Q : \mathbb{R}^2_+ \to [1, \infty)$ is continuous, twice continuously differentiable on $\mathbb{R}^2_{++}$, nondecreasing in both arguments, with $Q(0, P) = Q(R, 0) = 1$ and $\bar{Q}(R) \equiv \sup_P Q(R, P) < \infty$ for every $R$. Moreover $\ln Q$ is jointly concave and supermodular on $\mathbb{R}^2_+$.

Nothing is learned without reading or without a literature to read, and what one can learn is bounded by what one reads. Concavity of $\ln Q$ rules out increasing returns to reading. Supermodularity says that reading is worth more in a topic with more to learn from. Our running example is

$$Q(R, P) = \left(1 + \beta\, h(R, P)\right)^{\gamma}, \qquad h(R, P) = \frac{RP}{R + P}, \qquad \beta > 0,\ \gamma \in (0, 1]. \tag{2}$$

The harmonic aggregator $h$ is close to $R$ when the reader samples a small part of a rich literature and close to $P$ when she reads a thin literature exhaustively. Because $h$ is concave and homogeneous of degree one, the example satisfies Assumption 1 (Appendix A).

## 3.3 Reading protects

Each paper in the flow *covers* a given project independently with a small probability. The number of covering papers is therefore Poisson with mean $\theta P$. A covering paper is caught only if it is read.

With reading coverage $c = R/P$ it is caught with probability $\varphi(c)$.

**Assumption 2** (Detection)**.** $\varphi : \mathbb{R}_+ \to [0,1]$ is continuous and twice differentiable on $(0,\infty)$, with $\varphi(0) = 0$, $\varphi' > 0$ and $\varphi'' < 0$.

Concavity says that the first papers one reads are the most informative about overlap. Random reading with replacement gives $\varphi(c) = 1 - e^{-c}$. Our example $\varphi(c) = 1 - e^{-\sigma c}$ indexes *search precision* by $\sigma$: a better search engine finds covering papers sooner.

**Credit.** Overlap erodes credit multiplicatively. Each covering paper that goes unread turns part of the project into a duplicate and removes a share $\lambda$ of its credit. When a covering paper is caught, the researcher chooses between two options. She can *pivot* to a fresh variant at a time cost that leaves a share $1 - \kappa_p$. Or she can *proceed* as a follow-on that cites the earlier paper and keeps a share $\delta$ drawn from a distribution $G$. She proceeds if $\delta > 1 - \kappa_p$. The expected share kept after a detection is $1 - \ell \equiv \mathbb{E}\max\{\delta, 1 - \kappa_p\}$.

**Assumption 3** (Credit)**.** $0 \le \ell < \lambda \le 1$ and $\theta > 0$.

Detection is valuable ($\ell < \lambda$), and duplicates need not earn zero ($\lambda < 1$ is allowed). The arXiv evidence (Item 4) requires both. Let $\Delta \equiv \lambda - \ell$ and $q \equiv 1 - G(1 - \kappa_p)$, the share of detections that proceed.

**Lemma 1** (Credit and overlap)**.** *Under Assumptions 2 and 3, expected credit per project is*

$$C(R,P) = \exp\Big\{ -\theta P\big[\lambda - \Delta\,\varphi(R/P)\big]\Big\}. \tag{3}$$

*Covering papers that the researcher does not pivot away from (*flagged overlap*) arrive at rate* $\theta P\,[1 - (1-q)\varphi]$*. A share*

$$A = \frac{q\,\varphi}{1 - (1-q)\,\varphi} \tag{4}$$

*of them is* aware*: the later paper cites the earlier one. Unaware duplicates keep a share* $1 - \lambda$ *of credit. Aware follow-ons keep* $\mathbb{E}[\delta \mid \delta > 1 - \kappa_p]$*, which can exceed one.*

The lemma follows from Poisson thinning. Detected and undetected covering papers are independent Poisson counts with means $\theta P\varphi$ and $\theta P(1-\varphi)$, and $\mathbb{E}[t^K] = e^{-\mu(1-t)}$ for $K \sim \text{Poisson}(\mu)$. Three features matter below. Only $\theta\lambda$ and $\Delta/\lambda$ enter payoffs. Aware follow-ons are selected on high $\delta$, which is why they can out-earn comparable papers. And the measured overlap flag is a fixed share of covering events. Its level is a normalisation; its changes are not.

### 3.4 The researcher's problem

A researcher in a topic of density $P$ earns credit $a\,x\,Q(R,P)\,C(R,P)$ per period. Writing $x$ projects is worth $x$, and each is worth $QC$. Dividing by the common factor $a$,

$$g(r,P) \equiv (1-r)\,Q(br,P)\,C(br,P), \qquad v(P) \equiv \max_{r\in[0,1)} g(r,P). \tag{5}$$

The *payoff curve* $v$ is the central object of the paper. Writing speed $a$ scales every payoff and so drops out of $v$. The researcher takes $P$ as given: she ignores both that her papers teach her colleagues and that they add to what her colleagues must read.

### 3.5 The frontier, careers and catch-up

**Frontier.** Instead of joining an open topic, a researcher may explore new ground. An explorer opens a viable topic with probability $\pi(M)$, where $M$ is the measure of topics already open. If she succeeds, her output is worth $f_0$ per unit of writing speed: the founding papers of a field. The frontier pays $a\,F(M)$ in expectation, with $F(M) = f_0\,\pi(M)$. We assume $\pi$ is continuous, positive and nonincreasing, with elasticity $\eta \equiv -M\pi'(M)/\pi(M) \geq 0$: as more directions are taken, a fruitful new one is harder to find (Kortum, 1997; Bloom et al., 2020). Opening $M$ topics takes $\Phi(M) = \int_0^M \mathrm{d}m/\pi(m)$ researchers. We assume $F(M) < \hat{v}$, the peak of the payoff curve defined below, so open topics are worth joining. A newly opened topic attracts followers only if a thin topic beats the frontier, $F < v(P_0)$, where $P_0$ is the flow of a newborn topic.

**Careers and catch-up.** Researchers leave their topic at rate $\rho$ (retirement, a change of field) and are replaced by newcomers who choose a topic. The literature a newcomer must master is a stock $S$ that accumulates the flow and loses relevance at rate $\delta$: $\dot{S} = P - \delta S$, so $S = P/\delta$ in a stationary topic. A newcomer must read it at speed $b_c$ before contributing. This is the *catch-up* cost: a topic-level version of the burden of knowledge of Jones (2009). Explorers face no stock.

**Lemma 2** (Career value)**.** *With exponential careers, a newcomer who joins a stationary topic of density $P$ and then earns $a\,v(P)$ per period obtains, relative to an explorer, the expected career value $w(P)/F(M)$, where*

$$w(P) \equiv v(P)\,e^{-\kappa P}, \qquad \kappa \equiv \frac{\rho}{\delta\,b_c}. \tag{6}$$

*In a stationary topic a share $1 - e^{-\kappa P}$ of members are catching up, so $w(P)$ is also output per member and the topic has $n(P) = P\,e^{\kappa P}/(a\,x^*(P))$ members, where $x^*(P)$ is the optimal writing share.*

Catch-up takes $T = S/b_c = P/(\delta b_c)$ periods, and a career survives it with probability $e^{-\rho T} = e^{-\kappa P}$. The frictionless benchmark is $\kappa = 0$ ($b_c \to \infty$). Incumbents earn $a\,v(P)$ per period; newcomers earn $a\,w(P)$ on average over their careers.

### 3.6 Equilibrium

**Definition 1** (Topic equilibrium)**.** A stationary topic equilibrium is a density $P^*$ for every open topic, a writing share $x^*$ and a measure $M^*$ of open topics such that: (i) $x^* = x^*(P^*)$ solves (5); (ii) newcomers are indifferent between joining any open topic and exploring: $w(P^*) = F(M^*)$; (iii) the allocation is stable under entry: $w'(P^*) < 0$; (iv) researchers add up: $\Phi(M^*) + M^* n(P^*) = N$.

Condition (iii) selects allocations that survive the adjustment $\dot{n}_j \propto w(P_j) - F(M)$. At a crossing where $w$ rises, one more entrant raises the payoff of entering and draws in more. Topics in equilibrium need not have the same density a priori, but (ii) and (iii) leave a single admissible density, so they do. We state results in the *large-topic limit* $\Phi(M)/N \to 0$. Here topics are large relative to the researcher time needed to open one, as they are in the data, where pioneers are a small share of authors. Appendix A gives the general expressions.

## 4 The payoff curve

Every result in the paper rests on the shape of $v$. The draft version of this model could establish that shape only in examples. The next theorem shows it holds for every learning and detection technology that satisfies Assumptions 1 to 3.

**Assumption 4** (Interior reading and learning potential)**.** (a) Reading the first paper is worth its time: $b\,[\partial_R \ln Q(0,P) + \theta\Delta\varphi'(0)] > 1$ for all $P > 0$. (b) A thin literature is worth reading: for some $\varrho > 0$,

$$\nu_0(\varrho) \equiv \lim_{P\downarrow 0} \frac{Q(\varrho P, P) - 1}{P} - \frac{\varrho}{b} - \theta\big[\lambda - \Delta\varphi(\varrho)\big] > 0.$$

Part (b) compares, at the first papers of a topic, the gain from learning along the ray $R = \varrho P$ with the time spent reading and the overlap risk. In the example it reads $\sup_\varrho\{\gamma\beta\varrho/(1+\varrho) - \varrho/b - \theta[\lambda - \Delta(1 - e^{-\sigma\varrho})]\} > 0$.

**Theorem 1** (The payoff curve)**.** *Under Assumptions 1 to 3:*

*(i)* *$v$ is continuous on $\mathbb{R}_+$, $v(0) = 1$, $v(P) \geq e^{-\theta\lambda P} > 0$ and $\lim_{P\to\infty} v(P) = 0$.*

*(ii)* *$\ln v$ is concave on $(0,\infty)$. Under Assumption 4(a) it is strictly concave.*

*(iii)* *Under Assumption 4, there is a unique $\hat{P} \in (0,\infty)$ such that $v$ is strictly increasing on $[0,\hat{P}]$ and strictly decreasing on $[\hat{P},\infty)$, with $\hat{v} \equiv v(\hat{P}) > 1$. For every $F \in (0,\hat{v})$ the equation $v(P) = F$ has exactly one root above $\hat{P}$.*

*(iv)* *(Envelope) $v$ is differentiable on $(0,\infty)$ with*

$$\frac{v'(P)}{v(P)} = \underbrace{\partial_P \ln Q(R^*,P)}_{\text{more to learn from}} - \underbrace{\theta\big[\lambda - \Delta\big(\varphi(c^*) - c^*\varphi'(c^*)\big)\big]}_{\text{more to miss}}, \tag{7}$$

*where $R^* = b\,r^*(P)$ and $c^* = R^*/P$.*

*(v)* *The optimal reading share $r^*(P)$ is unique and nondecreasing in $P$. Papers per researcher, $a\,x^*(P)$, are nonincreasing in density.*

The proof (Appendix A) rests on one observation. $\ln g$ is *jointly* concave in reading time and density. $\ln(1-r)$ and $\ln Q$ are concave by assumption. The erosion exponent $-\theta\lambda P + \theta\Delta\,P\varphi(R/P)$

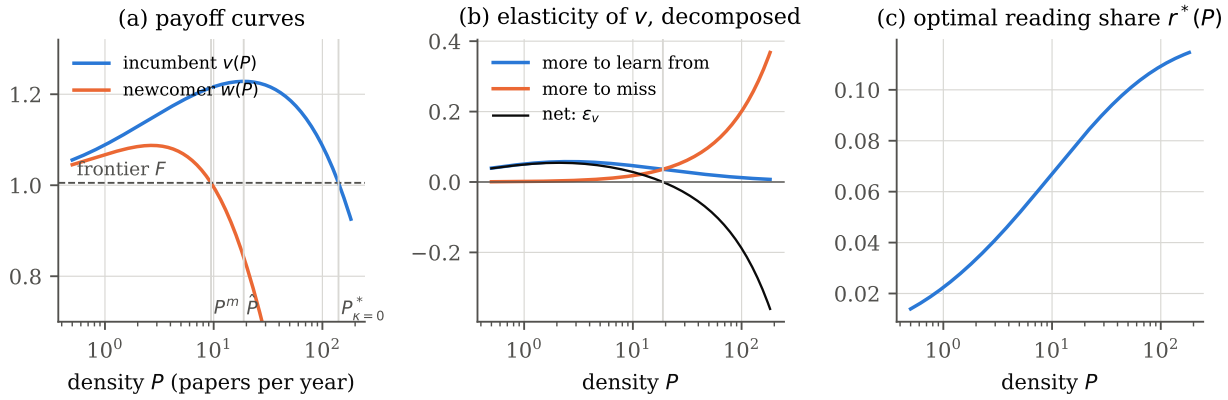



Figure 1: The payoff curve and its two forces. Left: $v(P)$ and the entrant's career payoff $w(P) = v(P)e^{-\kappa P}$ in the calibrated model, with the peak $\hat{P}$, the frictionless equilibrium (where $v = F$) and the mature density under catch-up costs (where $w = F$). Right: the decomposition (7) of $v'/v$ into the learning and overlap terms, and the optimal reading share.

is linear plus the perspective of the concave function $\varphi$, and perspectives preserve concavity. Maximising a jointly concave function over one argument leaves a concave function of the other, so $\ln v$ is concave. A log-concave function that starts at one, rises, and vanishes at infinity is single peaked. Part (v) follows from Topkis's theorem: $\partial^2 \ln C/\partial R\,\partial P = -\theta\Delta\varphi''(c)\,c/P \geq 0$ and $\ln Q$ is supermodular, so reading and density are complements.

The theorem answers the draft's first open problem ("find conditions on $Q$ and $\varphi$ that guarantee single-peakedness"). It needs no parametric assumption, and the conditions are economic: no increasing returns to reading, and diminishing returns to search. Note what the theorem does *not* need. Duplicates may keep most of their credit, since only $\lambda > \ell$ is required. The single peak comes from overlap that goes unread, not from an assumption that overlap is worthless.

**The sign of the spillover.** Let one more paper appear in a topic. By the envelope theorem, the change in every other member's payoff is exactly $a\,v'(P)$. The researchers' own reallocation of time is second order.

**Proposition 1** (Spillover sign)**.** *An additional paper in a topic of density $P$ changes the per-period payoff of each incumbent by $a\,v'(P)$. The change is positive for $P < \hat{P}$ (more to learn from) and negative for $P > \hat{P}$ (more to miss).*

The sign of the spillover is therefore *derived*, not assumed. A paper helps its topic when the topic is thin and hurts it when the topic is dense. Overproduction is not built in. Whether science is overcrowded depends on where equilibrium places topics on $v$. That is the subject of the next two sections.

## 5 Free topic choice without frictions

We first set $\kappa = 0$, so $w = v$. Newcomers move freely between topics and the frontier. This benchmark isolates the topic-choice margin and reproduces the draft's core claims, now as theorems.

**Proposition 2** (Equilibrium and dissipation)**.** *Let Assumptions 1 to 4 hold, $\kappa = 0$, and suppose $\hat{v} > F(N/n(\hat{P}))$. Then a topic equilibrium exists and is unique. Every open topic sits on the falling side of the payoff curve, $P^* > \hat{P}$, where an additional paper lowers the payoff of every member (Proposition 1). Payoff per researcher equals the frontier payoff $a\,F(M^*) < a\,\hat{v}$. The gain from cumulative knowledge, $\hat{v} - F(M^*)$, is competed away.*

The logic is that of the open-access fishery (Gordon, 1954; Weitzman, 1974), with one difference: the resource is attention to the literature, and the people who crowd it are the same people who must read it. Entry continues past the peak because a newcomer compares the *average* payoff in a topic with the frontier. She ignores that her papers lower everyone else's payoff once the topic is past $\hat{P}$. Crossings on the rising side are unstable: one more entrant raises the payoff of entering. So topics fill past the peak, to the crossing on the falling side.

**Volume opens new topics.** The next result is the paper's anchor. It says how a larger community distributes itself.

**Proposition 3** (Volume opens new topics)**.** *In the large-topic limit, equilibrium density depends on $(N, a)$ only through the product $Na$, and*

$$\mathcal{E} \equiv \frac{\mathrm{d}\ln P^*}{\mathrm{d}\ln N} = \frac{\eta}{\eta\, D + |\varepsilon_w|}, \qquad D \equiv 1 - \varepsilon_x + \kappa P^* \geq 1, \tag{8}$$

*where $\varepsilon_w = \mathrm{d}\ln w/\mathrm{d}\ln P < 0$ and $\varepsilon_x = \mathrm{d}\ln x^*/\mathrm{d}\ln P \leq 0$ are evaluated at $P^*$. New topics absorb a share $\mathrm{d}\ln M^*/\mathrm{d}\ln N = 1 - D\,\mathcal{E} = |\varepsilon_w|/(\eta D + |\varepsilon_w|)$ of growth, and papers per researcher move with elasticity $\varepsilon_x \mathcal{E} \leq 0$. If the frontier is perfectly elastic ($\eta = 0$), density, the overlap rate and the aware share do not depend on $N$, $a$ or the cost of entering science.*

Equation (8) holds for $\kappa \geq 0$; we use it again in Section 6. Three remarks.

First, the overlap rate is *pinned by entry, not by volume.* When the community grows, newcomers open new topics until existing ones are no more attractive than the frontier. The flagged overlap rate and the aware share are functions of $P^*$ alone (Lemma 1), so they inherit its invariance. A flat overlap rate while submissions double (Item 1) is what the model predicts. It is not evidence that crowding is harmless.

Second, a fixed field corresponds to $\eta \to \infty$. Then $\mathcal{E} \to 1/D$: growth thickens existing topics. The arXiv elasticity of 0.04–0.27, against a fixed-field benchmark of 0.53–0.89, implies that roughly three quarters of growth opens new ground. Inverting (8),

$$\eta = \frac{\mathcal{E}\,|\varepsilon_w|}{1 - \mathcal{E}\,D}, \tag{9}$$

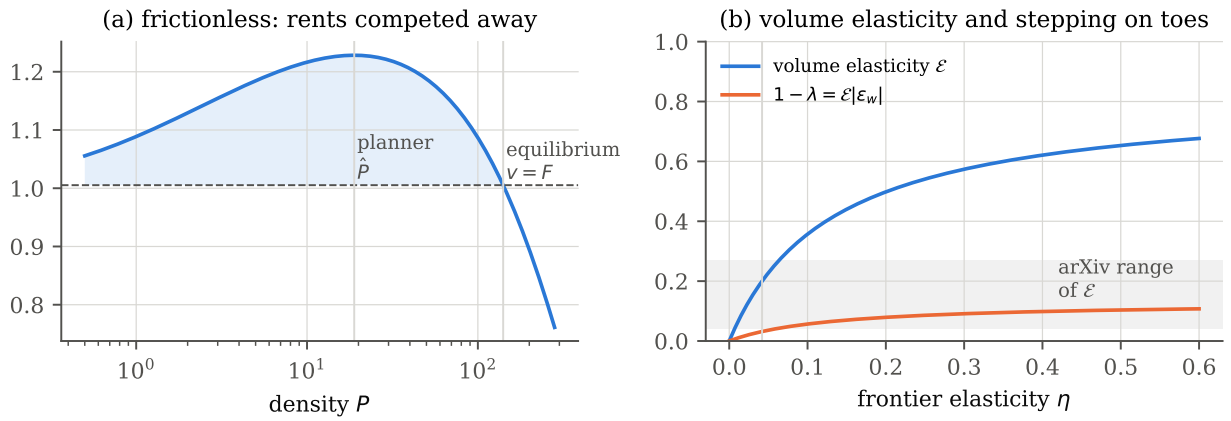



Figure 2: Left: without catch-up costs, topics fill to the density where $v = F$, and the shaded rent between the payoff curve and the frontier is competed away; the planner puts topics at the peak. Right: the volume elasticity $\mathcal{E}$ and the stepping-on-toes wedge $1 - \lambda = \mathcal{E}|\varepsilon_w|$ as functions of the frontier elasticity $\eta$, at the calibrated slope of $w$. The shaded band is the arXiv range of $\mathcal{E}$.

so the frontier elasticity is identified by the volume elasticity and the slope of the entrant's payoff curve (Fig. 2, right).

Third, papers per author barely move when the community grows, because $\varepsilon_x \mathcal{E}$ is the product of two small numbers. Growth through new authors with flat papers per author (Item 2) is the signature of cheaper entry, not of faster drafting (Section 7).

*Remark* 1 (Reading capacity per researcher or per field). The draft discussed a variant with a fixed screening capacity $B$ per field, so that coverage is $B/P$ and researchers write full time. Topic choice still pins density, so Propositions 2 and 3 survive. But if topics cannot split, overlap rises with field volume, which the arXiv volume test rejects. Per-researcher capacity also keeps the time-allocation margin that separates writing tools from reading tools (Section 7). We adopt it throughout.

**The planner.** A planner who controls how many researchers join each topic, but not how they split their time, maximises $Y = a\,[f_0 M + M\,n\,w(P)]$ subject to $\Phi(M) + Mn = N$.

**Proposition 4** (Overcrowding and the frontier subsidy)**.** *Let the conditions of Proposition 2 hold, with* $\kappa \geq 0$.

*(i) The planner's density* $P^o$ *satisfies*

$$\frac{w(P^o) - F(M^o)}{w(P^o)} = |\varepsilon_w(P^o)|\,\frac{1 + n(P^o)\,\pi(M^o)}{D(P^o)}, \tag{10}$$

*with* $\hat{P}_w < P^o < P^*$ *and* $M^o > M^*$, *where* $\hat{P}_w$ *is the peak of* $w$. *The planner opens more topics and puts fewer researchers in each. Overlap stays positive at the optimum. It is lower than in equilibrium whenever the credit lost to overlap rises with density, as it does in every calibration we report.*

*(ii) In the large-topic limit $P^o \to \hat{P}_w$. Topics sit at the peak of the entrant's payoff curve, which is the peak of $v$ when $\kappa = 0$.*

*(iii) A frontier subsidy (equivalently, a tax on joining open topics) of $s^o = w(P^o) - F(M^o)$ per unit of writing speed decentralises the planner's allocation.*

*(iv) (Two wedges) A planner who also chooses $x$ wants researchers to write less than $x^*(P)$ wherever $w'(P) < 0$ and more wherever $w'(P) > 0$. The writing wedge is proportional to $(\ln w)'(P)$. In the large-topic limit it vanishes at the optimal density, so controlling topic choice alone attains the first best.*

There are two wedges, and they differ in importance. Newcomers choose topics on the average payoff while the planner uses the marginal one: this is the *crowding wedge*. Inside each topic, researchers ignore that their writing adds to what colleagues must read: this is the *writing wedge*. Part (iv) shows the writing wedge is a symptom of the crowding wedge. Once topics are at their best density, researchers' private time allocation is efficient. Policies that work only inside topics are dissipated: writing caps, reading subsidies, better tools. With $\eta = 0$ the equilibrium payoff stays pinned at $aF$ whatever happens inside topics. Policies that change the frontier are not dissipated: frontier grants, credit for first papers in a cluster, prizes for opening fields. Fund the frontier, not the reading tools.

# 6 Catch-up costs: rents, overcrowding and topics that grow too fast

The frictionless benchmark predicts equal returns across topics. The arXiv data reject this (Item 3). The same researcher earns more citations in denser topics up to about one standard deviation above mean density, and less beyond it. We now let $\kappa > 0$: newcomers must read a topic's stock before contributing.

## 6.1 Mature topics

**Proposition 5** (Mature topics under catch-up costs)**.** *Let Assumptions 1 to 4 hold and $\kappa > 0$.*

*(i) $w(P) = v(P)e^{-\kappa P}$ is strictly log-concave, with a unique peak $\hat{P}_w < \hat{P}$.*

*(ii) In a topic equilibrium the mature density $P^m$ solves $w(P^m) = F(M)$ on the falling side of $w$. Incumbents earn a $v(P^m)$ per period. Their log premium over the frontier equals the catch-up cost, $\ln[v(P^m)/F(M)] = \kappa P^m = \rho T$, where $T = S/b_c$ is catch-up time.*

*(iii) Mature topics lie on the* rising *side of $v$ if and only if catch-up is costly relative to the rent at the peak: $\kappa\hat{P} > \ln(\hat{v}/F)$.*

*(iv) Let topics differ in fertility $\zeta$, so that payoffs are $\zeta v(P)$, and let $P^m(\zeta)$ solve $\zeta w(P) = F$.*

*Across mature topics, log returns rise with log density with slope exactly equal to the premium:*

$$\frac{\mathrm{d}\ln[\zeta\, v(P^m(\zeta))]}{\mathrm{d}\ln P^m(\zeta)} = \kappa P^m(\zeta) > 0.$$

Catch-up turns the dissipated rent into a protected one. Incumbents earn more than the frontier, and the premium exactly offsets what a newcomer gives up while reading the stock. More fertile topics attract more entrants, have larger stocks and so protect larger premia. This yields a positive returns-density gradient across topics, positive whether those topics lie left or right of $\hat{P}$. Part (iv) gives the model's reading of the 6.7% per standard deviation in Item 3. Compared across topics, the returns gradient measures the catch-up premium, not the slope of $v$.

### 6.2 Rising returns do not mean topics are too thin

A natural reading of Item 3 is that 88% of researcher-years sit where more density raises returns, so most topics are too thin. The model says otherwise.

**Proposition 6** (Overcrowding with rising returns)**.** *In every stable topic equilibrium, one more member in a topic of density $P$ changes total output of the other members by $n\, a\, w'(P)\, \mathrm{d}P/\mathrm{d}n < 0$. This counts both the spillover on incumbents ($v'$) and the extra catch-up she imposes on future newcomers ($-\kappa v$). Moreover:*

*(i) If $\kappa\hat{P} > \ln(\hat{v}/F)$, then $\hat{P}_w < P^m < \hat{P}$. Incumbents' returns rise with density ($v'(P^m) > 0$) and the topic is still too crowded ($w'(P^m) < 0$).*

*(ii) The planner's density is below $P^m$ in every topic (Proposition 4). The loss of output per researcher in the large-topic limit is $1 - F/\hat{w}$, where $\hat{w} = w(\hat{P}_w)$.*

The measured gradient is the incumbent's private return to density. A newcomer values a topic net of the reading she must do to join it, and so does the planner. Rising incumbent returns are the footprint of the catch-up rent, not of an untapped spillover. The welfare sign stays negative everywhere the entry margin is active. This replaces the hoped-for "two-sided" welfare conclusion with a sharper one: once the catch-up burden on future newcomers is counted, science is overcrowded at the margin even where crowding does not show up in returns. The only topics that are too thin are young ones that information frictions have not yet filled. We turn to these next.

### 6.3 When innovation is too fast

The stock lags the flow. A topic whose literature grows fast has a small stock relative to its current density. Newcomers find it cheap to join, so they keep joining after the point where a mature topic would close. This is the literal sense in which innovation can be "too fast".

Consider a topic born with flow $P_0$ and an empty stock. While entry pays, newcomers who learn about the topic join at a rate that makes its flow grow at the diffusion rate $g$: $P(t) = P_0 e^{gt}$ and $\dot{S} = P - \delta S$. A newcomer's career value relative to the frontier is $v(P)e^{-\rho S/b_c}/F$, so the topic

stops attracting newcomers at the first date $t_c$ at which $v(P(t_c))\, e^{-\rho S(t_c)/b_c} = F$. Call $P_c = P(t_c)$ the *stopping density*.

**Proposition 7** (Too fast)**.** *Suppose $w(P_0) > F$.*

*(i)* $P_c \geq P_c^{\infty}(g)$*, where $P_c^{\infty}(g)$ is the root on the falling side of*

$$v(P)\exp\Big(-\kappa\,\frac{\delta}{\delta+g}\,P\Big) = F.$$

*$P_c \to P_c^{\infty}(g)$ as $P_0 \to 0$. The function $P_c^{\infty}$ is continuous and strictly increasing, with $P_c^{\infty}(0) = P^m$ and $\lim_{g\to\infty} P_c^{\infty}(g) = \bar{P}_F$, the frictionless density where $v = F$.*

*(ii) If mature topics sit on the rising side of $v$ ($\kappa\hat{P} > \ln(\hat{v}/F)$), every topic that grows faster than*

$$g^* = \delta\left[\frac{\kappa\,\hat{P}}{\ln(\hat{v}/F)} - 1\right] = \frac{\rho\,\hat{P}}{b_c\ln(\hat{v}/F)} - \delta \tag{11}$$

*overshoots the peak: $P_c > \hat{P}$, so at the end of its growth spurt each extra paper lowers incumbents' payoffs. As $P_0 \to 0$, $g > g^*$ is also necessary.*

*(iii) After $t_c$, incumbents stay, since $v(P_c) > F$. Under the fast-stock approximation the topic thins at the turnover rate $\rho$ from $P_c$ to $P^m$. It spends $\ln(P_c/\hat{P})/\rho$ periods beyond the peak while consolidating.*

The proposition does three things. It locates the overcrowded researcher-years: they are in topics whose literature grew faster than $g^*$, and they persist long after the rush because incumbents never pay the catch-up cost that deters newcomers. It explains why the returns curve in Item 3 peaks and falls off at the top (Section 9 shows the calibrated life cycle reproduces it). And it identifies who sets the speed limit. The threshold $g^*$ falls with catch-up speed $b_c$, so literature-summarising tools that make catching up cheap make overshooting *more* common. It rises with turnover $\rho$ and with the peak density $\hat{P}$, and falls with the rent $\ln(\hat{v}/F)$.

## 7 Artificial intelligence as several different shocks

"AI" enters the model through several parameters, and each leaves a different footprint. That is what makes the model testable rather than a story.

**Proposition 8** (Signatures of AI)**.** *In the large-topic limit:*

*(i)* Faster drafting *($a \uparrow$) acts like a larger community. Density moves with elasticity $\mathcal{E}$ (zero if $\eta = 0$) and the number of topics with elasticity $1 - D\mathcal{E}$. Papers per researcher rise with elasticity $1 + \varepsilon_x\mathcal{E}$, and output per researcher with elasticity $1 - \mathcal{E}|\varepsilon_w|$.*

*(ii)* Cheaper entry into science *($c \downarrow$, with $N$ set by $aF(M) = \omega + c$ and $\eta > 0$) raises $N$. Density rises with elasticity $\mathcal{E}$ with respect to $N$, papers per researcher move with elasticity $\varepsilon_x\mathcal{E} \approx 0$, and new topics absorb a share $1 - D\mathcal{E}$ of growth.*

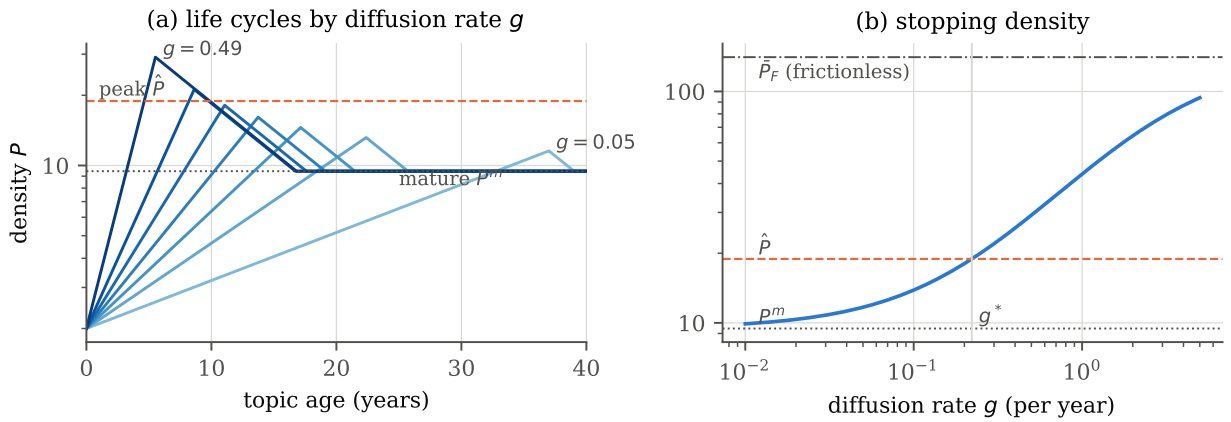



Figure 3: Topic life cycles. Left: density paths for topics with different diffusion rates $g$ in the calibrated model; topics growing faster than $g^*$ overshoot the peak $\hat{P}$ and consolidate slowly. Right: the stopping density $P_c^{\infty}(g)$ of Proposition 7, rising from the mature density $P^m$ toward the frictionless density $\bar{P}_F$.

*(iii)* Better reading, learning or search, or cheaper pivots *($b$, $\beta$ or $\sigma \uparrow$, $\kappa_p \downarrow$) raise $v$ and $w$ at every density. Density rises and the number of topics falls: science concentrates. Output per researcher rises only through the frontier, by $\eta\,|\mathrm{d}\ln M^*|$, and is unchanged when $\eta = 0$.*

*(iv)* Faster catch-up *($b_c \uparrow$, e.g. literature-summarising tools) raises $w$ at every density, with the effects in (iii). It also lowers the overshooting threshold $g^*$ of Proposition 7, so more topics grow too fast.*

*(v)* Better search *($\sigma \uparrow$) raises the aware share $A$ at any given reading coverage.*

Parts (iii) and (iv) follow from the envelope theorem. Each parameter raises $g(r, P)$ at every $(r, P)$, hence raises $v$. A pointwise upward shift of a single-peaked curve moves its falling-side crossing with a fixed $F$ to the right. The effect on the measured overlap rate is ambiguous in general: denser topics generate more overlap, but better detection removes more of it. Table 2 reports the calibrated signs.

**What the arXiv data say.** The footprints differ enough to tell the channels apart. Four facts settle the question for 2023–2026.

- Papers per author stayed flat while authors grew (Item 2). Faster drafting raises papers per author one for one. Cheaper entry leaves them flat.
- High-uptake areas produced *more* pioneer papers and *lower* local density relative to their own pre-ChatGPT trend (Item 6). All reading-side channels predict concentration: fewer new topics and denser old ones. Cheaper entry predicts more new topics. Faster growth also raises the share of young, thin topics in the life-cycle model, which lowers mean density.
- The aware share of close overlap stayed at 35–42% (Item 5). Better search would have raised

Table 2: The footprints of AI in the calibrated model

| Channel | Shock | Density | Topics | Papers/ researcher | Output/ researcher | Flagged overlap | Aware share (pp) | $g^*$ (pp) |
|---|---|---|---|---|---|---|---|---|
| Faster drafting | $a \times 1.3$ | +5.1 | +20.0 | +26.1 | +25.4 | +6.7 | -1.3 | +0.0 |
| Cheaper entry | $N \times 1.3$ | +5.1 | +20.0 | -0.1 | -0.8 | +6.7 | -1.3 | +0.0 |
| Faster reading | $b \times 1.3$ | +8.5 | -9.9 | +0.2 | +0.4 | +1.6 | +5.8 | +0.5 |
| Better learning | $\beta \times 1.3$ | +13.6 | -16.9 | -0.5 | +0.7 | +17.0 | -2.7 | -5.6 |
| Better search | $\sigma \times 1.3$ | +0.3 | -0.4 | +0.0 | +0.0 | -12.4 | +11.1 | +0.6 |
| Faster catch-up | $b_c \times 2$ | +61.3 | -61.3 | -1.5 | +2.6 | +79.5 | -13.6 | -23.6 |
| Cheaper pivots | $\kappa_p \div 2$ | +1.1 | -1.4 | -0.1 | +0.1 | -21.6 | -15.3 | +0.6 |

Long-run stationary effects in log points (×100) of a 30% shock (catch-up speed doubled, pivot cost halved). Density is $P^*$; output per researcher is $aF(M^*)$. The flagged overlap rate is relative to its baseline level, which is a normalisation (Section 2). $g^*$ is the overshooting threshold of Proposition 7.

it.

- Measured by meaning, close overlap did not fall (Item 7). Under cheaper entry it is pinned (Proposition 3).

So far AI has acted on science through *entry*. The reading channels remain predictions. When literature-summarising tools spread, the model predicts denser topics, fewer new ones, more overshooting by fast-growing topics, and little or no gain in output per researcher.

**Writing AI scales science; reading AI crowds it.** The asymmetry in Proposition 8 is the paper's sharpest policy implication. Writing speed multiplies every payoff, including the frontier's. Newcomers spread out, so a faster pen raises output almost one for one and leaves duplication unchanged. Reading speed improves only existing topics. Newcomers crowd in until those topics are no better than the frontier, so the improvement is competed away. This overturns a natural intuition. The worry that writing tools will flood science with duplicates is misplaced: free topic choice pins overlap. The hope that reading tools will raise productivity is also misplaced: free topic choice dissipates them. Tools are worth subsidising only if they make new topics easier to open, that is, if they raise $F$.

# 8 Aggregate implications

## 8.1 Stepping on toes, measured

Growth models since Jones (1995) let the flow of new ideas scale with researchers as $N^\lambda$, $\lambda < 1$, and call $1 - \lambda$ "stepping on toes". The parameter is reduced form. Here the flow of credited novel output is $Y = N\, a\, F(M^*(N))$, because in equilibrium every researcher's career payoff equals the frontier's. The model therefore gives $\lambda$ as an equilibrium object.

**Proposition 9** (Stepping on toes)**.** *In the large-topic limit,*

$$\lambda \equiv \frac{\mathrm{d}\ln Y}{\mathrm{d}\ln N} = 1 - \eta \frac{\mathrm{d}\ln M^*}{\mathrm{d}\ln N} = 1 - \mathcal{E}\,|\varepsilon_w(P^*)|. \tag{12}$$

*If, in addition, topics opened later are less fertile, with fertility elasticity* $\eta_f \geq 0$ *(Appendix B.1), the right-hand side of* (12) *is an upper bound on* $\lambda$*.*

The formula is a sufficient statistic built from two observable objects. The first is the volume elasticity of density, $\mathcal{E}$ (Test 1). The second is the elasticity of the entrant's payoff curve at equilibrium density, $\varepsilon_w = \varepsilon_v - \kappa P^*$. That is the incumbent's returns gradient minus the catch-up premium (Test 4 and Proposition 5). Crowding does not *amplify* frontier depletion, as the draft conjectured. It *buffers* it. The density margin absorbs part of the growth in researchers, so fewer new topics are opened and the frontier deteriorates more slowly. The cost of crowding shows up in the *level* of productivity, as the dissipated rent $\hat{w} - F$, and not in its elasticity. With $\mathcal{E} \approx 0.2$ and the calibrated slope of $w$, crowding and access depletion together imply $\lambda$ close to one (Section 9). The bulk of the measured decline in research productivity (Bloom et al., 2020) must come from elsewhere: falling fertility of new topics ($\eta_f$, not identified by density data) or selection on researcher ability, as in Ekerdt and Wu (2024).

### 8.2 A Solow paradox for science

The last half-century brought enormous improvements in access to the literature: online journals, preprint servers, full-text search, and now language models that read for us. Measured research productivity fell throughout (Bloom et al., 2020). Proposition 8(iii) suggests one reason.

**Proposition 10** (Reading improvements are dissipated)**.** *Let* $\omega$ *be any parameter that raises* $w$ *at every density and leaves* $F$ *unchanged. In the large-topic limit,* $\mathrm{d}P^*/\mathrm{d}\omega > 0$ *and* $\mathrm{d}M^*/\mathrm{d}\omega < 0$*, and*

$$\frac{\mathrm{d}\ln(Y/N)}{\mathrm{d}\omega} = \eta \left| \frac{\mathrm{d}\ln M^*}{\mathrm{d}\omega} \right| \geq 0,$$

*with equality if and only if* $\eta = 0$*. By contrast,* $\mathrm{d}\ln(Y/N)/\mathrm{d}\ln a = 1 - \mathcal{E}|\varepsilon_w|$*.*

Better access to knowledge raises the payoff of working in existing topics. Newcomers respond by crowding them until they are no better than the frontier. What remains is denser topics, fewer new ones, and the same output per researcher. The prediction matches Evans (2008), who finds that online access narrowed the set of papers that get cited. It also matches Hao et al. (2026), who find that AI tools "contract science's focus". Gains from access to frontier knowledge measured at the level of individuals or topics (Iaria et al., 2018; Biasi and Moser, 2021) are real, but in equilibrium they are partly competed away at the level of science.

# 9 Quantification

This section puts numbers on the mechanisms. The aim is not to fit every moment of the arXiv record. It is to check that one parameterisation of the model reproduces the main facts with plausible primitives, and to read off its implications for rents, overcrowding, AI and growth.

## 9.1 Strategy

We use the functional forms of Section 3: harmonic learning (2) and exponential detection. Time is measured in years. Table 3 lists the parameters. Some are set externally. The unit of density is fixed by $b = 100$ papers read per year at full-time reading. Careers leave a topic at $\rho = 0.10$ (ten-year tenure), the stock's relevance decays at $\delta = 0.25$ (a half-life of about three years), topics become obsolete at $\xi = 0.04$, and a newborn topic produces $P_0 = 2$ papers a year. The credit block comes from Item 4 through Lemma 1. With exponential follow-on credit and a pivot that costs 30% of a project, the +29% premium of aware follow-ons and the −19% penalty of unaware duplicates imply that 31% of detected overlaps proceed and that detection averts $\Delta/\lambda = 37\%$ of erosion. Search precision $\sigma$ is set so that the aware share at mature density equals the arXiv value of 38% (Item 5).

The remaining parameters are calibrated to Test 4 (Item 3): learning $(\beta, \gamma)$, erosion $(\theta\lambda)$, the frontier payoff $F$, catch-up speed $b_c$, and the distributions of diffusion rates and fertility. The targets are the gradient per standard deviation of density, the returns of the sparsest, fourth and densest fifths relative to the middle one, and the 12% of researcher-years beyond the peak. A soft target puts the incumbents' log premium $\kappa P^m = \rho T$ at about 0.15, a year and a half of catch-up per ten-year tenure. We simulate the stationary cross-section of topic life cycles (Proposition 7) and run the arXiv regressions on it. Researchers who move between topics supply the identifying variation in an author-fixed-effects design, and the model has no ability differences. The baseline analogue is therefore the cross-section of researcher-years. The within-topic analogue, which uses only life-cycle variation, is a robustness check. The search is a quasi-random scan of 8,192 points followed by Nelder–Mead from the best five (Appendix C).

We do not target the quadratic summary of Test 4. A quadratic in density imposes a symmetric hump. The model's payoff curve is asymmetric in log density, with a concave rise as learning saturates and an exponential fall as erosion accumulates. The arXiv fifths are asymmetric in the same way: −8%, +5%, +1%.

## 9.2 Fit

Table 4 and Fig. 4 compare model and data. The calibrated life cycle places the peak of the payoff curve 0.92 standard deviations above mean density, with 13% of researcher-years beyond it, against about one standard deviation and 12% in the data. Neither was imposed directly. The model matches the sparsest fifth (-7.1% against −8%) and the fourth fifth (3.5% against +5%). It produces a decline at the top, but a smaller one than the data: the densest fifth earns 6.7% against

Table 3: Parameters

| Parameter | Meaning | Value | Source |
|---|---|---|---|
| *Set externally* | | | |
| $b$ | papers read per year at full-time reading | 100 | normalisation (unit of $P$) |
| $\lambda$ | credit lost by an unaware duplicate | 0.19 | Fact 4 ($-19\%$) |
| $\kappa_p$ | credit lost by pivoting | 0.30 | assumption |
| $\Delta/\lambda$, $q$ | detection value; share proceeding | 0.37, 0.31 | Fact 4 ($+29\%$), Lemma 1 |
| $\sigma$ | search precision | 1.6 | aware share (Fact 5), see text |
| $\delta$ | decay of the stock's relevance | 0.25 | half-life $\approx 2.8$ years |
| $\rho$ | rate of leaving a topic | 0.10 | ten-year topic tenure |
| $\xi$ | topic obsolescence | 0.04 | 25-year topic life |
| $P_0$ | flow of a newborn topic | 2 | a few papers per year |
| *Calibrated (Table 4)* | | | |
| $\beta$, $\gamma$ | learning efficiency, curvature | 1.88, 0.131 | hump in returns |
| $\theta\lambda$ | erosion per earlier paper | 0.0020 | fall at the top |
| $F$ | frontier payoff | 1.005 | sparsest fifth |
| $b_c$ | catch-up reading speed | 20 | premium, share beyond peak |
| $g_{\text{med}}$, $s_g$ | diffusion rate: median, log-sd | 0.152, 0.74 | share beyond peak, fifths |
| $s_\zeta$ | log-sd of topic fertility | 0.075 | gradient across topics |
| $\eta$ | frontier elasticity | 0.0419 | $\mathcal{E} = 0.2$ (Fact 1), (9) |

$+1\%$. It also gives a gradient of 4.9% against 6.7%. Both misses have the same source. In the model the densest topics include the most fertile mature ones, whose incumbents earn high returns, alongside the overshooting topics that pull returns down. Author fixed effects in the data do not remove topic fertility either, so this composition effect is real. The data suggest the overshooting topics are more numerous or more crowded than our stylised life cycle allows. The within-topic calibration fits about equally well (last column).

Where are researchers in the calibrated cross-section? 45% of researcher-years are in mature topics, 25% in topics consolidating after a growth spurt, and 31% in growing topics. Overcrowded researcher-years sit overwhelmingly in consolidating topics that grew faster than $g^* = 22\%$ a year (Fig. 4, right).

### 9.3 What the calibrated model says

Table 5 collects the implications.

**Rents.** The payoff curve peaks at $\hat{P} = 19$ papers a year. There, building on others' work raises output by 9% relative to a newborn topic. Mature topics sit below the peak, at $P^m = 9.5$. Their incumbents earn a premium of 21% over the frontier, the reward for 1.9 years of catch-up. Without catch-up costs, topics would fill to the density where $v = F$ and the entire rent at the peak, 22% of the frontier payoff, would be dissipated. That is 18% of output per researcher.

**Overcrowding.** Even with rising incumbent returns ($\varepsilon_v(P^m) = 0.030 > 0$), the entrant's payoff curve is falling at the mature density ($|\varepsilon_w| = 0.159$). Mature topics are too crowded once the

Table 4: Fit to the arXiv facts

| Moment | arXiv | Baseline | Within-topic |
|---|---|---|---|
| Gradient per SD of density | 0.067 | 0.049 | 0.055 |
| Sparsest fifth vs. middle | -0.080 | -0.071 | -0.096 |
| Fourth fifth vs. middle | 0.050 | 0.035 | 0.023 |
| Densest fifth vs. middle | 0.010 | 0.067 | 0.028 |
| Researcher-years beyond the peak | 0.120 | 0.129 | 0.125 |
| Incumbents' log premium $\kappa P^m$ | 0.150 | 0.189 | 0.172 |
| Peak of $v$ (SD above mean)$^\dagger$ | 1.00 | 0.92 | 1.10 |
| Quadratic: linear term$^\dagger$ | 0.094 | 0.048 | 0.052 |
| Quadratic: squared term$^\dagger$ | -0.047 | -0.001 | -0.009 |
| Aware share of flagged overlap$^\dagger$ | 0.38 | 0.38 | 0.89 |

$^\dagger$ Not targeted. Baseline: cross-section of researcher-years (movers across topics). Within-topic: topic fixed effects, recalibrated. Density is the researcher-weighted standardised log flow.

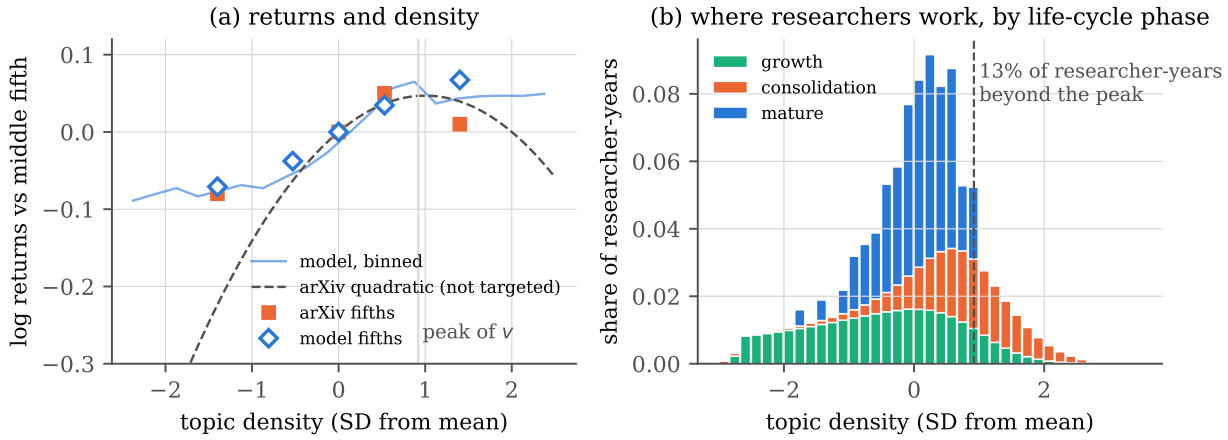



Figure 4: Test 4 in the model. Left: log returns by standardised topic density, relative to the middle fifth, in the calibrated cross-section, with the arXiv fifths and the arXiv quadratic (not targeted). Right: the distribution of researcher-years by density and life-cycle phase. Researcher-years beyond the peak are concentrated in topics consolidating after fast growth.

catch-up burden is counted (Proposition 6). Moving every topic to the peak of $w$ raises output per researcher by 8.2%. The optimal frontier subsidy is 8.2% of the frontier payoff. The gain is modest because catch-up costs already hold mature topics near the peak of $w$. Most of what free entry competes away is the rent of the frictionless benchmark.

**Growth.** Inverting (9) with $\mathcal{E} = 0.2$ gives a frontier elasticity of $\eta = 0.042$: the supply of new topics is close to perfectly elastic. The stepping-on-toes parameter is $\lambda = 1 - \mathcal{E}|\varepsilon_w| = 0.968$. Over the arXiv range $\mathcal{E} \in [0.04, 0.27]$ it lies between 0.994 and 0.957. Topic crowding and access depletion therefore explain only a small part of the diminishing returns to research found by Bloom et al. (2020). Their estimates require $\lambda$ well below one, which in our framework must come from fertility depletion (Appendix B.1) or from selection.

Table 5: Rents, overcrowding and growth in the calibrated model

| Object | Value |
|---|---|
| Peak of the payoff curve $\hat{P}$ (papers per year) | 18.9 |
| Gain from cumulative knowledge at the peak, $\hat{v}/v(P_0) - 1$ | 8.8% |
| Mature density $P^m$ | 9.5 |
| Catch-up time $T = S/b_c$ (years) | 1.89 |
| Incumbents' premium over the frontier, $e^{\kappa P^m} - 1$ | 20.8% |
| Rent at the peak, $\hat{v}/F - 1$ | 22.2% |
| Frictionless benchmark: rent dissipated, $1 - F/\hat{v}$ | 18.2% |
| Catch-up economy: planner's gain, $\hat{w}/F - 1$ | 8.2% |
| Optimal frontier subsidy, $s^o/F$ | 8.2% |
| Overshooting threshold $g^*$ (per year) | 0.221 |
| Slope of the entrant payoff curve $\lvert\varepsilon_w(P^m)\rvert$ | 0.159 |
| Frontier elasticity $\eta$ (from $\mathcal{E} = 0.2$) | 0.0419 |
| Stepping-on-toes parameter $\lambda = 1 - \mathcal{E}\lvert\varepsilon_w\rvert$ | 0.968 |

### 9.4 AI counterfactuals

Table 2 (in Section 7) and Fig. 5 report the long-run effects of 30% improvements in each channel. Faster drafting raises output per researcher by 25.4 log points: almost one for one, as Proposition 10 predicts. It raises papers per researcher by as much, and spreads researchers over more topics. Faster reading raises density by 8.5 log points and shrinks the number of topics, but raises output per researcher by only 0.41 log points: the dissipation of Proposition 10. Faster catch-up is the most powerful concentrating force. Doubling catch-up speed raises mature density by 61.3 log points and lowers the overshooting threshold $g^*$ by -23.6 percentage points, so many more topics grow too fast. Better search lowers flagged overlap and raises the aware share. That is the signature the arXiv data do not show (Item 5). Cheaper entry reproduces what they do show: more topics, flat papers per author and density nearly unchanged.

## 10 Held-out predictions

The catch-up extension was added after Test 4. The following predictions were not used to build it and can be tested with the project's existing files (cluster panels, the PDF citation sample, OpenAlex reference lists).

**P1.** *Overcrowding is young and fast.* The fall in returns beyond the peak should be concentrated in clusters whose flow grew faster than their stock, that is, with a high flow-to-stock ratio $P/S$ (Proposition 7). Among mature clusters the returns curve should be monotone. Test: interact density and density squared in the Test 4 regression with cluster age, or with $P/S$ computed from the cluster's own history.

**P2.** *Returns follow the stock, density follows the flow.* Holding flow density fixed, returns should rise with the cluster's stock of relevant literature (Proposition 5(ii): the premium is $\rho S/b_c$). Test: add the depreciated cumulative count of cluster papers to the Test 4 regression.

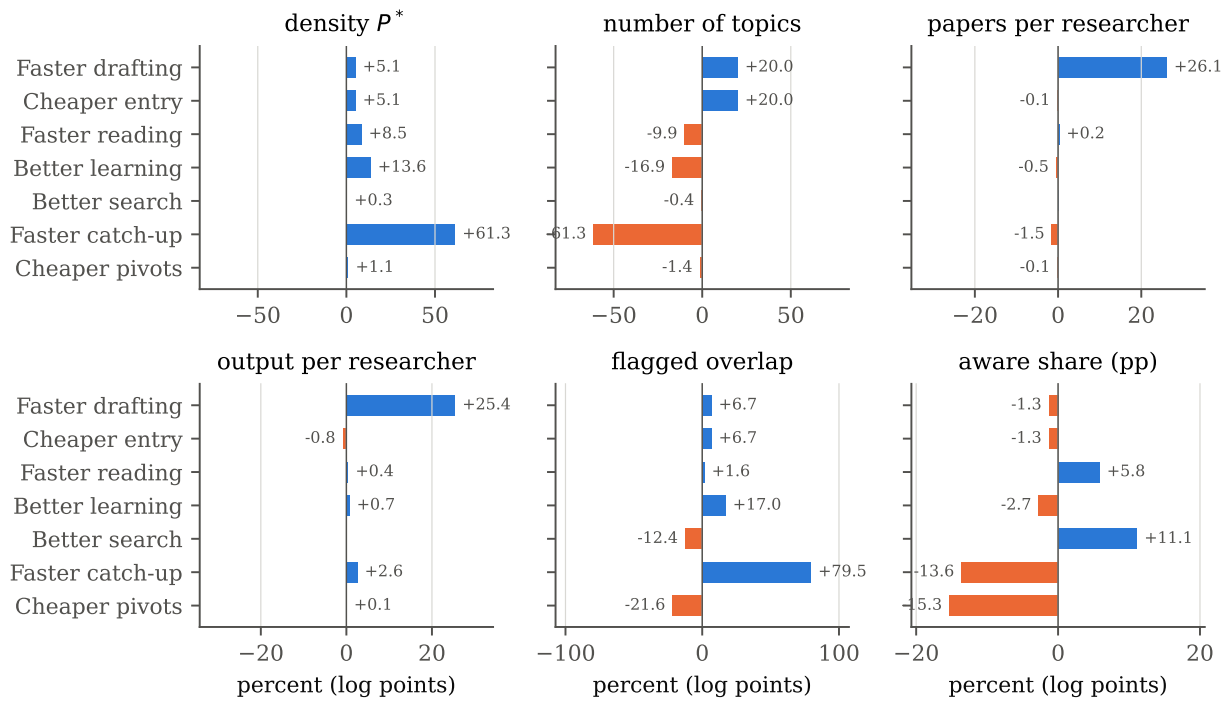



Figure 5: Long-run effects of AI channels in the calibrated model (30% shocks; catch-up speed doubled; pivot cost halved). Log points ×100; aware share in percentage points.

**P3.** *Newcomers start slowly where stocks are large.* The lag between a newcomer's arrival in a cluster and her first paper there, and the length of her first reference list, should rise with the cluster's stock. Incumbents should show no such gradient.

**P4.** *The cross-topic gradient is the catch-up premium.* The returns gradient estimated from researchers who *switch* clusters should equal the log premium of incumbents over pioneers in the clusters they join (Proposition 5(iv)). The within-cluster gradient over time should trace the payoff curve itself and peak at the same density.

**P5.** *When reading tools spread, science concentrates.* Once literature-summarising tools are widely used, high-uptake areas should form fewer new clusters, thicken existing ones, and show more overshooting by fast-growing clusters (Proposition 8(iii)–(iv)). Overlap measured by meaning should rise in dense clusters. Citations per author-year should not rise (Proposition 10).

**P6.** *Better search raises the aware share.* If search tools improve, the share of flagged pairs that cite the nearest earlier paper should rise, with no offsetting fall in the overlap rate in sparse clusters (Proposition 8(v)).

**P7.** *Close overlap is pinned; broad neighbourhoods fill.* Within-topic density is pinned by entry, while new topics are packed into an area's idea space. So the volume elasticity should *rise* with the similarity radius, from close to zero at the overlap radius toward the fixed-field benchmark at broad radii. Fact 1 already shows this ordering. The model predicts it should survive

embedding-based similarity.

**P8.** *Stepping on toes across areas.* Areas with a larger volume elasticity of density should show stronger diminishing returns to researchers in citation output, with $1 - \lambda = \mathcal{E}|\varepsilon_w|$ area by area (Proposition 9).

Two of the draft's original predictions are already settled. Equal returns (full dissipation) is rejected, and the catch-up extension replaces it. The search-channel reading of the 2026 drop in overlap is rejected because the drop is a wording artifact (Item 7). Both outcomes are consistent with the extended model.

## 11 Conclusion

Is innovation too fast? The model's answer is that the question is usually asked about the wrong margin. When papers arrive faster than they can be read, researchers do not keep piling into the same topics until duplication explodes. They spread out, and the overlap rate stays where entry pins it. The arXiv record bears this out: the volume of papers more than doubled while close overlap barely moved. The costs of speed lie elsewhere. The gains from building on others' work are competed away. Incumbents keep rents behind a wall of reading that newcomers must climb. Returns rise with density even though topics are too crowded at the margin. And topics whose literature grows faster than newcomers can catch up overshoot. Literally, they are too fast.

The policy implications follow from where the rents go. Anything that improves the payoff of working inside existing topics is competed away. That includes better reading tools, search engines, summarisers and caps on writing. Anything that improves the frontier is not. Writing tools scale science because they scale the frontier too. Reading tools crowd it unless they also make new topics easier to open. To date, artificial intelligence has changed science mainly by lowering the cost of entry, and the new entrants have spread out. The model predicts what will happen when tools that read for us become widespread: denser topics, fewer new ones, more hot topics that overshoot, and little gain in output per researcher. Unless, that is, the same tools are pointed at the frontier.

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# A Proofs

Throughout, "concave" means concave on the stated convex domain, and $\partial_R$, $\partial_P$ denote partial derivatives.

**The example satisfies Assumptions 1 and 2.** The function $h(R,P) = RP/(R+P)$ is half the harmonic mean of $R$ and $P$. It is concave, nondecreasing and homogeneous of degree one on $\mathbb{R}^2_+$, with $h \le \min\{R,P\}$. So $\ln(1+\beta h)$ is concave as a nondecreasing concave transformation of a concave function, and $Q$ is bounded by $(1+\beta R)^\gamma$ for fixed $R$. Supermodularity: $\partial_R \ln Q = \gamma\beta h_R/(1+\beta h)$ with $h_R = P^2/(R+P)^2$ and $h_{RP} = 2RP/(R+P)^3$, so

$$\partial_P \partial_R \ln Q = \gamma\beta \, \frac{h_{RP}(1+\beta h) - \beta h_R h_P}{(1+\beta h)^2} = \gamma\beta \, \frac{2RP(R+P) + \beta R^2 P^2}{(R+P)^4 (1+\beta h)^2} \ge 0.$$

For $\varphi(c) = 1 - e^{-\sigma c}$: $\varphi(0) = 0$, $\varphi' = \sigma e^{-\sigma c} > 0$, $\varphi'' < 0$.

**Proof of Lemma 1.** Covering papers form a Poisson count $K$ with mean $\theta P$. Each is detected independently with probability $\varphi$, so detected and undetected counts $K_d, K_u$ are independent Poisson with means $\theta P\varphi$ and $\theta P(1-\varphi)$. Credit is the product of $(1-\lambda)$ over undetected papers and of independent factors with mean $1-\ell$ over detected papers. Using $\mathbb{E}[t^K] = e^{-\mu(1-t)}$,

$$C = \mathbb{E}\big[(1-\lambda)^{K_u}\big]\,\mathbb{E}\big[(1-\ell)^{K_d}\big] = e^{-\theta P(1-\varphi)\lambda}\,e^{-\theta P\varphi\ell} = \exp\{-\theta P[\lambda-(\lambda-\ell)\varphi]\}.$$

Flagged overlap consists of undetected covering papers (rate $\theta P(1-\varphi)$) and detected ones that proceed (rate $\theta P\varphi q$). The aware share is the ratio of the second rate to the sum. Proceeding happens iff $\delta > 1-\kappa_p$, which gives the conditional mean. □

**Proof of Lemma 2.** Let careers end at rate $\rho$. A newcomer to a stationary topic reads for $T = S/b_c = P/(\delta b_c)$ periods and then earns $a\,v(P)$ until her career ends. Her expected career value is $\int_T^\infty e^{-\rho t} a\,v(P)\,\mathrm{d}t = a\,v(P)e^{-\rho T}/\rho = a\,w(P)/\rho$. An explorer's value is $aF(M)/\rho$. In a stationary topic with entry flow $e$, the mass of members in catch-up is $e(1-e^{-\rho T})/\rho$ and the mass writing is $e\,e^{-\rho T}/\rho$. So the writing share of members is $e^{-\kappa P}$, output per member is $a\,v(P)e^{-\kappa P}$, and (1) gives $P = a\,x^*(P)\,n\,e^{-\kappa P}$. □

**Proof of Theorem 1.** *(i)* Since $Q(0,P) = 1$ and $\varphi(0) = 0$, $v(P) \geq g(0,P) = e^{-\theta\lambda P}$. Because $Q$ and $\varphi$ are nondecreasing in $R$ and $R \leq b$,

$$g(r,P) \leq \bar{Q}(b)\exp\{-\theta P[\lambda - \Delta\varphi(b/P)]\} \xrightarrow[P\to\infty]{} 0,$$

since $\varphi(b/P) \to 0$ and $\lambda > 0$. At $P = 0$, $g(r,0) = 1-r$, so $v(0) = 1$. Define $\ln C(R,0) = 0$. Then $|\theta P\varphi(R/P)| \leq \theta P$ gives joint continuity of $g$ on $[0,1]\times\mathbb{R}_+$ ($g \to 0$ as $r \to 1$). Berge's maximum theorem gives continuity of $v$.

*(ii)* On $D = [0,1)\times(0,\infty)$,

$$\ln g(r,P) = \ln(1-r) + \ln Q(br,P) - \theta\lambda P + \theta\Delta\,P\varphi\left(\frac{br}{P}\right).$$

The first term is concave in $r$. The second is concave by Assumption 1 composed with a linear map. The third is linear. In the fourth, $(R,P) \mapsto P\varphi(R/P)$ is the perspective of the concave function $\varphi$, which is jointly concave on $\mathbb{R}\times\mathbb{R}_{++}$ (Boyd and Vandenberghe, 2004, §3.2.6). So $\ln g$ is jointly concave on $D$. Since $\ln g \to -\infty$ as $r \to 1$, the supremum over $r$ is attained in $[0,1)$. Partial maximisation of a jointly concave function over a convex set preserves concavity (Boyd and Vandenberghe, 2004, §3.2.5). Hence $\ln v(P) = \max_r \ln g(r,P)$ is concave.

For strict concavity, take $P_1 \neq P_2$ with maximisers $r_1, r_2$ and $t \in (0,1)$. Under Assumption 4(a), $\partial_r \ln g(0,P) > 0$, so $r_i > 0$. If $r_1 \neq r_2$, strict concavity of $\ln(1-r)$ gives $\ln g(r_t,P_t) > t\ln g(r_1,P_1) + (1-t)\ln g(r_2,P_2)$. If $r_1 = r_2 = r > 0$, then $P \mapsto P\varphi(br/P)$ has second derivative $\varphi''(br/P)(br)^2/P^3 < 0$, which gives the same strict inequality. In either case $\ln v(P_t) \geq \ln g(r_t,P_t) >$

$t \ln v(P_1) + (1-t) \ln v(P_2)$.

*(iii)* Fix $\varrho$ with $\nu_0(\varrho) > 0$. For small $P$, $r = \varrho P/b$ is feasible and

$$\frac{v(P)-1}{P} \geq \frac{g(\varrho P/b, P)-1}{P} = \frac{(1-\varrho P/b)\, Q(\varrho P, P)\, e^{-\theta P[\lambda - \Delta\varphi(\varrho)]} - 1}{P} \longrightarrow \nu_0(\varrho) > 0.$$

So $v > 1 = v(0)$ on some interval $(0, \epsilon)$. A strictly log-concave function is strictly quasi-concave, so its maximiser is unique. It is positive because $v$ exceeds $v(0)$ near zero, and finite because $v \to 0$. Strict quasi-concavity gives strict monotonicity on either side of $\hat{P}$. Since $v$ falls continuously and strictly from $\hat{v}$ to 0 on $[\hat{P}, \infty)$, each $F \in (0, \hat{v})$ has exactly one root there.

*(iv)* For $P > 0$ the maximiser $r^*(P)$ is unique (strict concavity in $r$) and $\partial_P g$ is continuous. The envelope theorem (Milgrom and Segal, 2002) gives $v'(P) = \partial_P g(r^*(P), P)$. Differentiating $\ln C$,

$$\partial_P\big[-\theta\lambda P + \theta\Delta P \varphi(R/P)\big] = -\theta\lambda + \theta\Delta\big[\varphi(c) - c\varphi'(c)\big], \qquad c = R/P,$$

which is (7). Concavity and $\varphi(0) = 0$ give $\varphi(c) - c\varphi'(c) \geq 0$.

*(v)* $\partial_r \partial_P \ln g = b\,[\partial_R \partial_P \ln Q + \partial_R \partial_P \ln C]$. The first term is nonnegative by Assumption 1. For the second, $\partial_R \ln C = \theta\Delta\varphi'(R/P)$, so $\partial_P \partial_R \ln C = -\theta\Delta\varphi''(c)\, c/P \geq 0$. So $\ln g$ has increasing differences in $(r, P)$, and the unique maximiser is nondecreasing by Topkis's monotonicity theorem (Topkis, 1998). □

**Proof of Proposition 1.** By Theorem 1(iv), the effect of a marginal paper on an incumbent's per-period payoff is $a\, v'(P)$. Its sign is that of $\hat{P} - P$ by Theorem 1(iii). □

**Proof of Proposition 2.** With $\kappa = 0$, $n(P) = P/(ax^*(P))$ is continuous and strictly increasing, since $x^*$ is nonincreasing by Theorem 1(v). Let $M(P)$ solve $\Phi(M) + Mn(P) = N$. It is unique, continuous and strictly decreasing in $P$ ($\Phi$ is increasing). Define $\Psi(P) = \ln v(P) - \ln F(M(P))$ on $[\hat{P}, \infty)$. $\ln v$ is strictly decreasing there, and $F(M(P))$ is nondecreasing because $F$ is nonincreasing, so $\Psi$ is strictly decreasing. In the large-topic limit $M(\hat{P}) = N/n(\hat{P})$ and $\Psi(\hat{P}) > 0$ by assumption. As $P \to \infty$, $v \to 0$ while $F(M(P)) \geq F(M(\hat{P})) > 0$, so $\Psi < 0$ eventually. Hence there is a unique root $P^* > \hat{P}$. Crossings on $(0, \hat{P})$ violate stability, since there $v' > 0$. Payoff per researcher is $a\, v(P^*) = aF(M^*) < a\hat{v}$. □

**Proof of Proposition 3.** In the large-topic limit $M = N/n(P) = N\, a\, x^*(P) e^{-\kappa P}/P$, so the equilibrium condition $w(P) = F\big(Nax^*(P)e^{-\kappa P}/P\big)$ involves $N$ and $a$ only through $Na$. Log-differentiating with respect to $\ln N$,

$$\varepsilon_w\, \mathrm{d}\ln P = -\eta\,[\mathrm{d}\ln N - D\,\mathrm{d}\ln P], \qquad D = \frac{\mathrm{d}\ln n}{\mathrm{d}\ln P} = 1 - \varepsilon_x + \kappa P.$$

Solving gives (8). Then $\mathrm{d}\ln M = \mathrm{d}\ln N - D\,\mathrm{d}\ln P$, and papers per researcher $a\, x^*(P)$ move with elasticity $\varepsilon_x\, \mathcal{E}$. If $\eta = 0$, $P^*$ solves $w(P) = \bar{F}$, which involves neither $N$ nor $a$. The overlap rate and

the aware share are functions of $P^*$ by Lemma 1. Free entry into science changes $P^*$ only through $N$. The general (non-limit) version replaces $D$ by $D\,(1-s_e)$ and $|\varepsilon_w|$ by $|\varepsilon_w|[1+s_e(\phi_e-1)]$, where $s_e=\Phi(M)/N$ and $\phi_e=M\Phi'(M)/\Phi(M)$. □

**Proof of Proposition 4.** Write $D=1-\varepsilon_x+\kappa P$; as above $n\,\mathrm{d}P/\mathrm{d}n=P/D$. The Lagrangian $a[f_0M+Mn\,w(P(n))]-\mu[\Phi(M)+Mn-N]$ has first-order conditions

$$\mu=a\,w\Big(1+\frac{\varepsilon_w}{D}\Big)\quad(n),\qquad a(f_0+nw)=\mu\Big(\frac{1}{\pi}+n\Big)\quad(M),$$

using $\Phi'(M)=1/\pi(M)$. Eliminating $\mu$ and using $F=f_0\pi$ gives (10). Along the resource constraint, $\mathrm{d}n/\mathrm{d}M=-(1/\pi+n)/M$, so

$$\frac{\mathrm{d}Y}{\mathrm{d}M}=a(f_0+nw)-a\,w\Big(1+\frac{\varepsilon_w}{D}\Big)\Big(\frac{1}{\pi}+n\Big).$$

At the equilibrium ($w=F=f_0\pi$, $\varepsilon_w<0$) this equals $-a\,f_0(\varepsilon_w/D)(1+n\pi)>0$. The planner therefore opens more topics, and density falls below $P^*$. At $\hat{P}_w$ ($\varepsilon_w=0$) it equals $af_0(1-w/F)<0$ whenever $w(\hat{P}_w)>F$. So the optimum lies in $(\hat{P}_w,P^*)$. For (ii), the left side of (10) is bounded by $(\hat{w}-F)/\hat{w}$ while $1+n\pi\to\infty$, so $|\varepsilon_w(P^o)|\to 0$ and $P^o\to\hat{P}_w$ by strict log-concavity of $w$. For (iii), with a subsidy $s$ to explorers the equilibrium condition becomes $w(P)=F(M)+s$, and $s=s^o$ makes $(P^o,M^o)$ satisfy it. For (iv), let the planner also set $x$. Per-member output is $a\,g(x,P)e^{-\kappa P}$, and at fixed $n$, $\partial\ln P/\partial\ln x=1/(1+\kappa P)$. The first-order condition for $x$ is

$$\partial_x\ln g+[\partial_P\ln g-\kappa]\frac{P}{x(1+\kappa P)}=0.$$

At the private optimum $\partial_x\ln g=0$ and, by the envelope theorem, $\partial_P\ln g-\kappa=(\ln w)'(P)$. The wedge therefore has the sign of $(\ln w)'$. In the large-topic limit the planner maximises $\max_{x,P}g(x,P)e^{-\kappa P}=\max_P w(P)$. The maximum is at $(x^*(\hat{P}_w),\hat{P}_w)$, where the wedge is zero. □

**Proof of Proposition 5.** (i) $\ln w=\ln v-\kappa P$ is strictly concave. Since $(\ln w)'(\hat{P})=-\kappa<0$, the peak of $w$ lies below $\hat{P}$. (ii) The equilibrium condition and stability are those of Definition 1. The premium follows from $v(P^m)e^{-\kappa P^m}=F$ and $\kappa P=\rho T$. (iii) $P^m<\hat{P}$ iff $w(\hat{P})<F$, since $w$ is falling on $(\hat{P}_w,\infty)$; this reads $\ln\hat{v}-\kappa\hat{P}<\ln F$. (iv) From $\ln\zeta+\ln v(P^m)=\ln F+\kappa P^m$, differentiate with respect to $\ln P^m$. □

**Proof of Proposition 6.** Total member output in a topic is $n\,a\,w(P(n))$, so one more member changes the output of the others by $n\,a\,w'(P)\,\mathrm{d}P/\mathrm{d}n$. Here $\mathrm{d}P/\mathrm{d}n>0$ and $w'<0$ at a stable equilibrium. Writing $w'=(v'-\kappa v)e^{-\kappa P}$ separates the spillover on incumbents from the catch-up burden. Part (i) combines Proposition 5(iii) with stability. Part (ii) is Proposition 4: per-researcher output is $aF$ in equilibrium and tends to $a\hat{w}$ at the optimum. □

**Proof of Proposition 7.** On the growth path, $S(t) = \int_0^t P_0 e^{gs} e^{-\delta(t-s)} \mathrm{d}s = P(t)\,\frac{1-e^{-(g+\delta)t}}{g+\delta} < \frac{P(t)}{g+\delta}$. Hence the entry payoff satisfies $v(P)e^{-\rho S/b_c} > w_g(P) \equiv v(P)e^{-\kappa\delta P/(\delta+g)}$. The function $w_g$ is strictly log-concave. Since $w_g(P_0) \geq w(P_0) > F$, the set $\{P \geq P_0 : w_g(P) > F\}$ is an interval $[P_0, P_c^\infty(g))$. Entry therefore continues at least until $P$ reaches $P_c^\infty(g)$, so $P_c \geq P_c^\infty(g)$. As $P_0 \to 0$, $t_c \to \infty$ and $S/P \to 1/(g+\delta)$, so $P_c \to P_c^\infty(g)$. Monotonicity: $w_g$ is strictly increasing in $g$ at every $P > 0$, so the falling-side root increases. The limits follow from $w_0 = w$ and $w_\infty = v$. (ii) $P_c^\infty(g) > \hat{P}$ iff $w_g(\hat{P}) > F$, that is, $\ln \hat{v} - \kappa\delta\hat{P}/(\delta+g) > \ln F$, which rearranges to $g > g^*$. Combined with (i), $g > g^*$ implies $P_c > \hat{P}$, and the converse holds in the limit $P_0 \to 0$. (iii) At $t_c$, $v(P_c)e^{-\rho S_c/b_c} = F$ implies $v(P_c) > F$, so incumbents do not leave. With no entry, membership decays at rate $\rho$. Under the fast-stock approximation $S = P/\delta$, entry resumes when $w(P) = F$, that is, at $P^m$. □

**Proof of Proposition 8.** (i) and (ii) restate Proposition 3: $P^*$ depends on $Na$; output per researcher is $aF(M^*)$; papers per researcher are $a\,x^*(P^*)$. Under free entry into science $F(M^*) = (\omega+c)/a$. For (iii)–(iv), $\partial_b \ln g = r[\partial_R \ln Q + \partial_R \ln C] \geq 0$. $Q$ is increasing in $\beta$. $C$ is increasing in $\sigma$ and decreasing in $\ell$, and $\ell$ is increasing in $\kappa_p$. $w$ is decreasing in $\kappa = \rho/(\delta b_c)$. So $w$ rises pointwise, which moves the falling-side root of $w(P) = F$ to the right. $M^* = N/n(P^*)$ falls because $n$ is increasing. $g^*$ in (11) is decreasing in $b_c$. (v) $A$ in (4) is increasing in $\varphi$, and $\varphi$ is increasing in $\sigma$. □

**Proofs of Propositions 9 and 10.** In the large-topic limit every researcher's career payoff is $aF(M^*)$, so $Y = NaF(M^*)$ and $\mathrm{d}\ln Y/\mathrm{d}\ln N = 1 - \eta\,\mathrm{d}\ln M^*/\mathrm{d}\ln N$. By Proposition 3, $\mathrm{d}\ln M^*/\mathrm{d}\ln N = |\varepsilon_w|/(\eta D + |\varepsilon_w|)$, so $\eta\,\mathrm{d}\ln M^*/\mathrm{d}\ln N = \mathcal{E}|\varepsilon_w|$. The bound with fertility depletion is proved in Appendix B.1. For Proposition 10, $Y/N = aF(M^*)$ and $M^*$ falls with $\omega$ by Proposition 8(iii). For writing speed, $P^*$ depends on $Na$, so $\mathrm{d}\ln(Y/N)/\mathrm{d}\ln a = 1 - \eta\,\mathrm{d}\ln M^*/\mathrm{d}\ln(Na) = 1 - \mathcal{E}|\varepsilon_w|$. □

# B Extensions

## B.1 Fertility depletion

Suppose topics also differ in fertility, and later topics are less fertile. The $m$-th topic opened has fertility $f(m) = \bar{f}\,m^{-\eta_f}$, so members of topic $m$ earn $af(m)$ times the payoff curve. Explorers succeed with probability $\pi(M) = \bar{\pi}M^{-\eta_a}$, which captures *access depletion*: elasticity $\eta_a$, the $\eta$ of the main text. A successful explorer opens a topic of fertility $f(M)$. Newcomers are indifferent between topic $m$ and exploring when

$$f(m)\,w(P_m) = \pi(M)f(M) \iff w(P_t) = \bar{\pi}\,M^{-\eta_a}\,t^{\eta_f}, \qquad t \equiv m/M \in (0,1].$$

**Proposition 11** (Scale invariance under fertility depletion)**.** *(i) If $\eta_a = 0$, the distribution of*

*densities across topics, and hence of overlap rates and aware shares, is invariant to $N$ and $a$, whatever the value of $\eta_f$. (ii) In general, with $\mathcal{D} \equiv \mathrm{d}\ln M^*/\mathrm{d}\ln N$,*

$$\lambda = 1 - (\eta_a + \eta_f)\,\mathcal{D}, \qquad \mathcal{D} = \Big(1 + \eta_a\,\mathbb{E}_n\Big[\frac{D(P_t)}{|\varepsilon_w(P_t)|}\Big]\Big)^{-1},$$

*where $\mathbb{E}_n$ averages over topics with weights proportional to members. The paper-weighted volume elasticity of density identifies $\eta_a$, not $\eta_f$, and $\lambda \le 1 - \eta_a\mathcal{D}$, the value obtained when $\eta_f = 0$. With a single density, $1 - \eta_a\mathcal{D} = 1 - \mathcal{E}|\varepsilon_w|$.*

*Proof.* (i) With $\eta_a = 0$ the indifference condition pins $P_t$ as a function of the relative rank $t$ alone. Members per topic $n(P_t)$ are then invariant, so $M = N/\int_0^1 n(P_t)\mathrm{d}t$ scales with $N$. (ii) Every researcher's career payoff is $a\pi(M)f(M)$, so $Y = Na\bar{\pi}\bar{f}M^{-\eta_a-\eta_f}$ and $\lambda = 1 - (\eta_a + \eta_f)\mathcal{D}$. At a fixed rank, $\varepsilon_w(P_t)\,\mathrm{d}\ln P_t = -\eta_a\,\mathrm{d}\ln M$. Substituting into $N = M\int_0^1 n(P_t)\mathrm{d}t$ gives $\mathrm{d}\ln N = \mathrm{d}\ln M\,(1 + \eta_a\mathbb{E}_n[D/|\varepsilon_w|])$. □

Fertility depletion, the textbook "fishing out" of ideas (Kortum, 1997), thus leaves the anchor result intact and strengthens it: volume opens new topics exactly. It adds to stepping on toes a component that density data cannot see. This is why Proposition 9 delivers an upper bound on $\lambda$.

### B.2 Fertility heterogeneity and the cross-section

With fertility $\zeta$ drawn independently at a topic's birth, a mature topic solves $\zeta w(P^m) = F$. More fertile topics are denser, their incumbents earn more, and the cross-sectional returns gradient is $\kappa P^m$ (Proposition 5(iv)). Topics with $\zeta\hat{w} < F$ cannot sustain replacement entry. They shrink once their growth spurt ends and are eventually abandoned. This is the model's version of research lines that die out.

### B.3 Heterogeneous ability

Ekerdt and Wu (2024) attribute part of measured stepping on toes to falling researcher ability. Let ability $z$ multiply a researcher's credit. If ability and reading are complements ($Q$ increasing in $z\,R$), able researchers gain more from dense literatures and sort into them. If ability also raises the pioneer's success probability, they sort to the frontier. Which force wins determines whether the dense-topic premium in Item 3 partly reflects sorting. Author fixed effects remove level differences in ability but not ability-specific returns to density. The two can be separated by interacting density with pre-period ability (early-career citations). We leave this for the empirical paper.

### B.4 Credit rules

Partial credit enters only through $\Theta = \theta\lambda$ and $\Delta/\lambda$. Theorem 1 requires $\lambda > \ell$ and nothing else. The sign of the externality at a stable equilibrium (Propositions 2 and 6) therefore survives whenever duplicates earn some credit. Replication value can be added as a positive social value of

aware follow-ons that is not credited privately. This raises the social value of detection relative to the private one, but it does not change the sign of $w'$ at equilibrium.

### B.5 Transitional dynamics after a writing shock

A one-time increase in $a$ raises every topic's flow immediately, before membership adjusts. Topics jump to the right along $v$, overlap rises, and incumbents' payoffs fall. If the jump lands beyond $\hat{P}$, the shock is crowding. Newcomers then go to the frontier and topics thin through turnover, returning to $P^*$ at rate $\rho$. The overlap rate spikes and decays with half-life $\ln 2/\rho$. Papers per author rise immediately and stay higher. Flat papers per author and flat overlap by meaning (Items 2 and 7) rule out a large writing shock in 2023–2026.

## C Numerical methods and verification

**Solving the researcher's problem.** For each $P > 0$ the first-order condition $\partial_r \ln g(r, P) = 0$ has a unique root in $[0, 1)$ because $\ln g$ is strictly concave in $r$. We find it by vectorised bisection (60 steps) followed by three Newton steps, and set $r^* = 0$ when $\partial_r \ln g(0, P) \leq 0$. The derivative $v'/v$ is computed from the envelope formula (7), not by differencing. Peaks and crossings use bounded Brent searches on the log-concave curves.

**Stationary equilibrium and planner.** The equilibrium solves $\ln w(P) = \ln F(M(P))$ on $[\hat{P}_w, \infty)$. Here $M(P)$ solves $\Phi(M) + Mn(P) = N$, and the left side minus the right side is strictly decreasing (proof of Proposition 2). The constrained planner maximises output over $P$ along the resource constraint. The full planner maximises over $(P, x)$ by Nelder–Mead.

**Life cycles.** The growth phase has closed forms, $P = P_0 e^{gt}$ and $S = P(1 - e^{-(g+\delta)t})/(g + \delta)$. The stopping time is found on a fine time grid and refined by linear interpolation. Consolidation and maturity follow Proposition 7(iii). Diffusion rates are lognormal and discretised into equal-probability bins. Fertility is lognormal and discretised the same way. Topic-years are weighted by members, $P/x^*(P)$, and by topic survival, $e^{-\xi t}$.

**Verification.** The file `code/tests/test_propositions.py` checks every analytical result numerically:

- log-concavity, single-peakedness, $v(0) = 1$ and $v \to 0$ on 25 random parameter draws satisfying Assumptions 1 to 4;
- the envelope formula against finite differences, and the monotonicity of $r^*(P)$ (Theorem 1);
- the initial slope $v'(0^+) = \sup_\varrho \nu_0(\varrho)$ (Assumption 4(b));
- the externality decomposition and its sign (Proposition 1);

- existence, uniqueness and the falling-side location of equilibrium for several $(\kappa, \eta)$, and dissipation (Proposition 2);
- the volume elasticity (8) against finite differences, and exact invariance to $N$ and $a$ when $\eta = 0$ (Proposition 3);
- the planner's first-order condition, the ordering $\hat{P}_w < P^o < P^*$, the large-topic limit and the vanishing writing wedge (Proposition 4);
- the cross-topic returns gradient $\kappa P^m$ (Proposition 5) and overcrowding with rising returns (Proposition 6);
- the monotonicity and limits of the stopping density, the threshold $g^*$, and $P_c \geq P_c^{\infty}(g)$ for finite birth flows (Proposition 7);
- dissipation of reading improvements (Proposition 10) and the stepping-on-toes formula (12) against finite differences (Proposition 9);
- a Monte Carlo check of the credit formula (3) (Lemma 1).

All checks pass.